

One curve, or a family of twists

How `ec-padic-closure`, `ec-density-bm` and `wild-symbols` sit against `kummer-padic-density.typ` and the survey — including one duplication, one collapse, and one correction

checked in `density-bridge.gp`, output in `results/density-bridge.txt`

Duplication, stated up front. `ec-padic-closure.typ` re-derives §2.1 of `kummer-padic-density.typ`. The density criterion, the index $M = c_p \cdot \#\tilde{E}^{\text{ns}}(\mathbb{F}_p)$, the E_1/E_2 test read off the denominator of $x(Q)$, and the “ $\tilde{E}(\mathbb{F}_p)$ non-cyclic forces rank ≥ 2 ” obstruction are all already there. I did not check the existing documents before writing it. What is genuinely new in it is narrow: the conductor-89 witness where the reduction map is **surjective** and density still fails, the census of the component-group layer, and the phrasing of the criterion as surjectivity modulo p^2 .

1. The criterion was already right here

`kummer-padic-density.typ` §2.1 states, for $\Gamma \subseteq E_d(\mathbb{Q})$:

$$\bar{\Gamma} = E_d(\mathbb{Q}_p) \iff \begin{cases} \Gamma \twoheadrightarrow E(\mathbb{Q}_p)/E_1(\mathbb{Q}_p) & (\text{index } M) \\ \Gamma \cap E_1 \not\subseteq E_2 \end{cases}$$

with $M = c_p \cdot \#\tilde{E}^{\text{ns}}(\mathbb{F}_p)$. That is exactly the N_p and the $v_p(c(mP)) = 1$ test of `ec-padic-closure.typ`.

Worth noting for its own sake: **the second clause is precisely the one that the phrasing “ $E(\mathbb{Q}) \rightarrow E(\mathbb{F}_p)$ is surjective” drops.** The correction made in that document was not a correction to this repository — the correct statement was already in it. The conductor-89 curve at $p = 11$ and $p = 13$ is just an explicit witness that the second clause has content.

The CM table row reproduces exactly. For $d = 6$, in the bad class $[u \cdot 3]$:

twist	curve	Kodaira	c_3	$M = c_3(3 - a_3)$
$d = 6$ (class $[u \cdot 3]$)	$y^2 = x^3 - 432$	IV^*	3	9
$d = 3$ (class $[3]$)	$y^2 = x^3 - 54$	IV^*	1	3

— the Tamagawa jump that forces rank ≥ 2 in one class and not the other.

2. The collapse: why the single-curve argument says nothing on X

This is the substantive difference, and it is worth being blunt about.

In `ec-density-bm.typ` the obstruction pairs **a point against a class**: $\text{inv}_v \mathcal{A}_\beta(Q) = \langle Q_v, \beta_v \rangle_v$ with β running over $H^1(\mathbb{Q}, E)$. On the Kummer surface that shape is not available. A point of X over the class δ is an unordered **pair** (P, Q) of points on one twist E_d , so what reciprocity has to work with is

$$\langle \delta_v P, \delta_v Q \rangle_v,$$

both entries in the local Kummer image $W_v = \text{im}(E_d(\mathbb{Q}_v)/n)$.

And W_v is maximal isotropic — $W_v^\perp = W_v$, the fact used in `ec-density-bm.typ` §7 to identify which classes can obstruct. So $\langle \delta_v P, \delta_v Q \rangle_v = 0$ **identically, at every place**, and reciprocity gives $0 = 0$. The single-curve Brauer–Manin argument is not merely weaker in the family setting; on X it is **vacuous**.

$\dim W_v$ is exactly half of $\dim H^1(\mathbb{Q}_v, E[2])$ at every place — checked for the conductor-37 curve at $v = 3, 5, 7, 11, 23, 37$. That is the whole of the collapse.

2.1. What rescues it

kummer-padic-density.typ §5.1.5 twists the pairing by a **non-scalar** $\varphi \in \text{End}_G(E[n])$:

$$\beta_v(P, Q) = \langle \delta_v P, \varphi \delta_v Q \rangle_v.$$

φ is Galois-equivariant, so reciprocity still gives $\sum_v \beta_v(P, Q) = 0$ — but W_v carries no isotropy constraint for the **twisted** form, so the sum is not forced to be $0 = 0$. If $\beta_v \equiv 0$ for every $v \neq p$, the image of $E_d(\mathbb{Q})$ in W_p is β_p -isotropic, hence of dimension ≤ 1 in a non-degenerate symplectic plane — which is exactly the measured failure to span.

The price is a hypothesis with no analogue in the single-curve story:

$E[n]$ must be decomposable. Otherwise $\text{End}_G(E[n]) = \mathbb{F}_n$, every φ is scalar, and β collapses back to the pairing that vanishes on W_v . In the CM family this is supplied by the $j = 0$ geometry: $\psi_3(x) = 3x(x^3 + 4k)$ with $k = -2d^3$ makes $-4k = (2d)^3$ a perfect cube, so a **second** rational root appears and $E[3] = C_1 \oplus C_2$ for the whole family.

Note what this means for §5 of `ec-density-bm.typ`: there I showed an index-2 obstruction on a curve with **no rational isogeny at all**. In the family setting that curve would be useless — decomposable $E[n]$ is not a convenience there, it is the hypothesis. The two settings want opposite things from $E[n]$.

3. What `wild-symbols.typ` actually contributes

Not a new tool — `level3.gp` already built the wild cubic symbol at 3. What the computation adds is **sharpness** of the lemma that makes it work.

`level3.gp` evaluates the wild symbol by the product formula, which needs every class of $K^\times / (K^\times)^3$, $K = \mathbb{Q}_3(\zeta_3)$, to have a global representative; its stated reason is that $U^{(4)} \subseteq (K^\times)^3$, so a class is fixed by its valuation and its unit part **modulo 9**. Brute force over $\mathcal{O}/\pi^{12} = \mathcal{O}/3^6$ confirms this and shows the 4 cannot be lowered:

m	units $\equiv 1 \pmod{\pi^m}$	all cubes?
2	59049	no (52488 fail)
3	19683	no (13122 fail)
4	6561	yes
5	2187	yes

At $m = 3$ exactly one third of the units are cubes — one full layer of the filtration would be lost. **The mod-9 precision in `level3.gp` is exactly right and cannot be reduced.**

4. A correction to `ec-density-bm.typ` §6

That section called the index-11 case “blocked by an unimplemented wild symbol”. That is too pessimistic, and this repository already contains two ways round.

(a) The product-formula evasion generalises. `level3.gp` needs only (i) a single prime above ℓ in $\mathbb{Q}(\zeta_\ell)$, and (ii) a depth bound giving global representatives. Condition (i) is automatic — ℓ is totally ramified in $\mathbb{Q}(\zeta_\ell)$ for every ℓ — and (ii) is the depth computation of `wild-symbols.typ`:

field	h	primes above ℓ	e	depth $pe/(p-1)$
$\mathbb{Q}(\zeta_3)$	1	1	2	3
$\mathbb{Q}(\zeta_{11})$	1	1	10	11

At 11 the depth is 11, so $U^{(12)}$ consists of 11-th powers; and $v(11^2) = 20 \geq 12$, so a class is pinned down modulo 121. The two inputs that made the cubic symbol computable are both present.

(b) **Often you do not need to evaluate the symbol at all.** The survey’s treatment of 11a1 at $p = 11$ was flagged as needing the quintic symbol and then did not use one: the proof needs β_{11} to be **non-degenerate**, not to be computed, and split multiplicative reduction settles that structurally. The Tate μ_5 turns out to be a **third** line, distinct from both global subgroups C_1, C_2 , so φ restricts to an isomorphism and the Weil pairing of two distinct lines of a symplectic plane is perfect.

So “no explicit class” and “no proof” are different things, and the survey had already separated them. What `wild-symbols.typ` establishes is narrower than §6 suggested: the **cost** of evaluating a wild symbol — depth $pe/(p-1)$, in a field ramified by $p-1$, on a class group of size p^{p+1} — not the impossibility of getting the theorem.

5. The K3 Brauer group, and why the search stops at $\ell = 5$

The recollection is **right in substance and wrong as stated**, and the repair is worth having because it makes the survey’s criterion into a Brauer-group statement exactly.

5.1. The literal claim is false

For a K3 surface over an algebraically closed field of characteristic 0,

$$\mathrm{Br}(\overline{X}) \cong (\mathbb{Q}/\mathbb{Z})^{22-\rho}, \quad \rho = \mathrm{rk} \, \mathrm{NS}(\overline{X}) \leq 20,$$

so $\mathrm{Br}(\overline{X})[\ell] \cong (\mathbb{Z}/\ell)^{22-\rho}$ is non-zero for **every** ℓ . The geometric Brauer group of a K3 never runs out of ℓ -torsion.

5.2. The true statement, and it is the criterion

The finiteness lives one level up, in the **Galois-invariant** part. For $A = E_1 \times E_2$ the Künneth decomposition of H^2 gives

$$\mathrm{Br}(\overline{A})[\ell] \cong \mathrm{Hom}(E_1[\ell], E_2[\ell]) / (\mathrm{Hom}(E_1, E_2) \otimes \mathbb{F}_\ell),$$

the quotient being the algebraic (Néron–Severi) part. For $E_1 = E_2 = E$ without CM, $\mathrm{Hom}(E, E) = \mathbb{Z}$, so the quotient is by the **scalars**:

$$\mathrm{Br}(\overline{A})[\ell] \cong \mathrm{End}(E[\ell]) / \mathbb{F}_\ell \cong M_2(\mathbb{F}_\ell) / \text{scalars},$$

and $\mathrm{Br}(\overline{\mathrm{Kum}(A)}) \cong \mathrm{Br}(\overline{A})$ away from 2 (Skorobogatov–Zarhin). Taking Galois invariants:

$$\mathrm{Br}(\overline{X})^{G_\ell}[\ell] \cong \mathrm{End}_G(E[\ell]) / \mathbb{F}_\ell, \quad X = \mathrm{Kum}(E \times E).$$

So the survey’s criterion $\mathrm{End}_G(E[\ell]) \neq \mathbb{F}_\ell$ is **literally** “the Kummer surface has a Galois-invariant transcendental Brauer class of order ℓ ”. Searching for the twisting endomorphism φ and searching for ℓ -torsion in that group are the **same search** — and the scalars are divided out precisely because they are the algebraic part, which is exactly why the criterion asks for φ **non-scalar**.

5.3. What actually bounds ℓ : Mazur–Kenku, not K3 geometry

$\mathrm{End}_G(E[\ell]) \neq \mathbb{F}_\ell$ forces the mod- ℓ image to be small. Over $\mathbb{Q}$, for **odd** ℓ , only one of the four cases in the survey’s triage table can occur:

structure	what it costs over $\mathbb{Q}$
two stable lines (split Cartan)	a non-cuspidal rational point of $X_{\mathrm{split}(\ell)} \cong X_0(\ell^2)$. By Mazur–Kenku the cyclic isogeny degrees over $\mathbb{Q}$ are $N \leq 19$ together with

	$\{21, 25, 27, 37, 43, 67, 163\}$; $N = \ell^2$ therefore forces $\ell^2 \in \{4, 9, 25\}$, i.e. $\ell \in \{2, 3, 5\}$. Note $\ell^2 = 49$ is not on the list.
one stable line, equal characters	$\chi_1 = \chi_2$ with $\chi_1 \chi_2 = \chi_{\text{cyc}}$ gives $\chi_1^2 = \chi_{\text{cyc}} \bmod \ell$; with a rational ℓ -torsion point ($\chi_1 = 1$) this needs $\ell = 2$.
image inside a nonsplit Cartan	impossible over $\mathbb{Q}$ for odd ℓ . Complex conjugation c lies in the image with $\det c = \chi_{\text{cyc}(c)} = -1$ and $c^2 = 1$, so its eigenvalues are $1, -1$, distinct for odd ℓ . An element of a nonsplit Cartan acts with eigenvalues α, α^ℓ for $\alpha \in \mathbb{F}_{\ell^2}$ — equal if $\alpha \in \mathbb{F}_\ell$, conjugate and outside $\mathbb{F}_\ell$ otherwise — never $\{1, -1\}$. So the fourth row of the triage table is empty over $\mathbb{Q}$.
larger irreducible image	$\text{End}_G = \mathbb{F}_\ell$, scalar.

So over $\mathbb{Q}$ the usable case is $\ell \in \{2, 3, 5\}$, exactly. A brute-force census agrees, with a caveat: **ellisomat** records isogeny targets up to **isomorphism**, so two distinct stable lines with isomorphic quotients register once — **14a1** has isogeny degrees $\{1, 2, 3, 6\}$ yet ψ_3 has **two** rational roots, hence two stable 3-lines. The census is therefore a lower bound on which ℓ occur, not a proof; it returns $\ell \in \{2, 3, 5\}$ over 44741 curves, and the exact statement rests on Mazur–Kenku. And $\ell = 7$ genuinely fails even where one would most expect it — the CM curve of conductor 49, $j = -3375$ with CM by $\sqrt{-7}$, has isogeny degrees $\{1, 2, 7, 14\}$: a **single** 7-isogeny, so $E[7]$ is reducible but not decomposable and $\text{End}_G(E[7])$ is scalar.

Verdict. Yes, there is a connection, and it is exact rather than analogical: the search for a twisting endomorphism is the search for ℓ -torsion in $\text{Br}(\overline{X})^{G_\mathbb{Q}}$. But the bound is not a K3 fact — a K3’s geometric Brauer group has ℓ -torsion for every ℓ . It is the vanishing of the **Galois-invariant** part, and that vanishing is Mazur–Kenku on rational points of modular curves. The mechanism runs out at $\ell = 5$ for the same reason there are no rational ℓ^2 -isogenies past 25.

6. Does β_p vary with the twist?

No — it depends on d only through its square class at p . So one check per class suffices: four at an odd p , eight at 2, not one per twist. But the shortcut that seems to follow does not, and the reason is worth separating out.

6.1. Why β_p is twist-independent within a class

If d/d' is a square in $\mathbb{Q}_p$ then

$$\psi : E_d \longrightarrow E_{d'}, \quad (x, y) \mapsto (c^2 x, c^3 y), \quad c^2 = d/d',$$

is an isomorphism **defined over $\mathbb{Q}_p$** . It carries $E_d[\ell]$ to $E_{d'}[\ell]$, each Galois-stable line to the corresponding one — so it commutes with φ — and preserves the Weil pairing. Hence it identifies W_p with $W_{p'}$ and β_p with $\beta_{(p)'}$. The same form, on the nose.

Checked on **14a1** ($E[3]$ decomposable: ψ_3 has two rational roots), $\ell = 3$, critical place $p = 7$. Every twist in the class [1] — the 7-adic units $d \equiv 1, 2, 4$ — gives identical local data, and a different class gives different data:

	class at 7	d tested	Kodaira	c_7	a_7
[1]		1, 2, 4, 8, 9, 11, 15, 22, 23, 37, 247, −6, −5	I_3	3	+1
non-square		3, 5, 6, 10, 13	I_3	1	−1

$M = c_7(7 - a_7)$ is 18 throughout the first row and 8 throughout the second.

6.2. What does move with d

The vanishing **away** from p . β_v can be non-zero only where $\dim W_v = 2$, which needs $\mu_\ell \subset \mathbb{Q}_v$, i.e. $v \equiv 1 \pmod{\ell}$. So the places at risk are

$$\{p\} \cup \{\ell\} \cup \{q \mid d : q \equiv 1 \pmod{\ell}\},$$

and that last set genuinely moves. Among the same 14a1 twists, $d = 1, 2, 4, 11, 15, 23$ contribute nothing new, while $d = 37$ brings in 37 and $d = 247$ brings in 13 and 19 — each a fresh place where β_v could survive and let the reciprocity sum balance away from the critical place. That is exactly the family-uniform obligation, and it is why the CM table in `kummer-padic-density.typ` lists “few bad primes necessary” as one of the four things the mechanism explains.

6.3. Why the shortcut still fails

$\beta_p \neq 0$ is a **local** fact about a form on W_p . “ $E_d(\mathbb{Q})$ fails to surject onto W_p ” is a **global** fact about one twist. The implication runs one way only:

$$\beta_p \neq 0 \text{ and } \beta_v = 0 (v \neq p) \implies \text{image isotropic} \implies \dim \leq 1 \implies \text{no surjection}.$$

It cannot be inverted. A twist can fail to surject simply because its rank is too small, which says nothing whatever about β_p . So testing one twist — or seven hundred — gives evidence, not proof; the 708 rank- ≥ 2 twists of the CM search are exactly that, and `kummer-padic-density.typ` presents them as exactly that.

There **is** a valid converse use, in the refuting direction. If a single twist in the class **does** span W_p (with $\beta_v = 0$ away from p), then β_p vanishes on W_p and the mechanism is dead for that class. So one twist can kill the obstruction; no number of twists can establish it.

Net effect on the mental picture. The symbol work, where it is needed at all, is needed **once per square class at the critical place** — a bounded, local computation, not something that recurs across an infinite family. And often it is not needed even once: 11a1 at $p = 11$ gets non-degeneracy structurally from the Tate curve. The genuinely d -dependent labour is the **vanishing** at the other places, and that is done by the dimension count $\dim W_v \leq 1$, which needs no symbol at all.

7. Dictionary

	single curve	twist family on X
what is asked	is $E(\mathbb{Q})$ dense in $E(\mathbb{Q}_p)$?	is $X(\mathbb{Q})$ dense in $X(\mathbb{Q}_p)$?
reduces to	one closure	4 (or 8) twists, one per square class
local criterion	the same one: onto $E(\mathbb{Q}_p)/E_1$, and meet E_1 outside E_2	ditto, per twist
what pairs	a point against a class $\beta \in H^1(\mathbb{Q}, E)$	a point against a point — both in W_v
plain reciprocity	gives the obstruction	gives $0 = 0$: W_v is Lagrangian
fix	none needed	twist by non-scalar $\varphi \in \text{End}_G(E[n])$
needs	nothing special about $E[n]$	$E[n]$ decomposable
obstruction is	a linear functional	a pairing — so no preferred line, which is why the search for a universal functional failed

The short version. The local half of the single-curve work was already in `kummer-padic-density.typ`. The global half — Brauer–Manin as summed local Tate pairings — is the right frame, but on the Kummer surface it degenerates to $0 = 0$ and only the **twisted** pairing survives, which is why the family argument needs decomposable $E[n]$ and the single-curve argument does not. And the wild-symbol material describes the cost of an explicit class, not a barrier to the theorem: `level3.gp` paid that cost at 3, and the survey showed at 11 that it often need not be paid at all.

Companion file: `density-bridge.gp`, run as

```
gp -q -s 12000000000 density-bridge.gp < /dev/null > results/density-bridge.txt
```

It reproduces the CM Tamagawa row at $p = 3$, the half-dimension count showing W_v Lagrangian, the sharpness of $U^{(4)} \subseteq (K^\times)^3$ by brute force over $\mathcal{O}/3^6$, and the cyclotomic data at 3 and 11 behind Section 4.